

Workshop Exercises — Crossover to Work Outcomes

Companion exercises for the exercise dyadic dataset (`exercise_data.RData`). This dataset is **independent** from the original Hahn-et-al-style dataset used by scripts 02–08. The exercise scenario studies how work-related psychological states of one partner (affect, self-determination, job crafting) crossover to influence the other partner’s work outcomes (engagement, performance, creativity), and how three dyad-level moderators (live_together, years_together, time_spent_this_morning_together) shape those pathways.

Dataset summary

Element	Specification
N couples	250 (500 individuals)
Predictors (person-level)	<code>affect</code> , <code>sdt</code> , <code>job_crafting</code>
Outcomes (person-level)	<code>engagement</code> , <code>performance</code> , <code>creativity</code>
Crossover moderators (dyad-level)	<code>live_together</code> (binary), <code>years_together</code> (continuous), <code>time_spent_this_morning_together</code> (continuous)
Distinguishing variable	<code>gender</code> (male / female)
Within-dyad predictor correlations	~0.30 (<code>affect</code> , <code>sdt</code> , <code>jc</code>)
Within-dyad outcome correlations	~0.25 (<code>engagement</code> , <code>performance</code> , <code>creativity</code>)
Partner predictors in long format	<code>partner_affect</code> , <code>partner_sdt</code> , <code>partner_job_crafting</code>
Wide format	<code>_a</code> / <code>_p</code> suffixes (actor / partner)

Focal moderated-crossover effects (one per outcome)

Outcome	Focal partner predictor	Focal moderator	DGP interaction
engagement	<code>partner_job_crafting</code>	<code>live_together</code>	<code>partner_jc</code> × <code>live_together</code> = 0.15
performance	<code>partner_sdt</code>	<code>years_together</code>	<code>partner_sdt</code> × <code>years_together</code> = 0.015
creativity	<code>partner_affect</code>	<code>time_spent_this_morning_together</code>	<code>partner_affect</code> × <code>time_spent</code> = 0.003

Files

File	Purpose
<code>exercises/simulate_exercise_data.RDGP</code>	script (run once to regenerate <code>exercise_data.RData</code>)
<code>exercises/exercise_data.RData</code>	Contains <code>ddl2</code> (long) and <code>ddw2</code> (wide)
<code>exercises/exercises.md</code>	This file
<code>exercises/answers_template.R</code>	Step-by-step instructions and code stubs

How to use these exercises

The exercises assume you have the exercise dataset loaded and the workshop's reference scripts (02–08) available for syntax. Each section below describes the goal and what to produce. Open `exercises/answers_template.R` for the step-by-step instructions and code stubs. After producing each output, write a one-sentence interpretation before moving on — that articulation is the exercise.

Section 1 — Data simulation & inspection

What you are practising: orienting yourself to a new dyadic dataset before any modelling.

What to produce: Read the header of the simulation script to understand the data-generating parameters for the three outcomes and the focal moderated-crossover effects. Run the script to create the data file. In a fresh R session, load the data and inspect both the long and wide formats. For each of the three outcomes, fit a null multilevel model and compute the ICC to assess within-dyad dependence. Summarise the three dyad-level moderators to understand their distributions. Compute within-dyad correlations for the three predictors to verify the simulated coupling. Run paired t-tests on the predictors and outcomes to identify which show non-zero gender mean differences.

Reflection prompt: Which of the three outcomes shows the largest gender difference in mean, and which shows the smallest? Does the difference match the DGP intercept gap, or is it diluted by the gender-asymmetric predictors?

Section 2 — Indistinguishable dyads: multilevel modelling

What you are practising: the basic Actor-Partner Interdependence Model in long format, treating dyad members as exchangeable.

What to produce: Load the long-format data. For each of the three outcomes, fit the full Actor-Partner Interdependence Model with all three actor

predictors, all three partner predictors, the three dyad-level moderators, and a random intercept for dyad. Also fit a reduced model that omits the three partner predictors, and compare the two to test whether the partner block is jointly needed. Repeat the process for all three outcomes. For each outcome, compute profile-likelihood confidence intervals and write a two-sentence interpretation of the actor and partner effects.

Reflection prompt: For which outcome is the partner block (partner_affect, partner_sdt, partner_job_crafting) jointly most important? Does this match the data-generating process?

Section 3 — Indistinguishable dyads: SEM

What you are practising: three structural equation modelling paradigms for indistinguishable dyads and the formal indistinguishability test.

What to produce: Load the wide-format data. For each of the three outcomes, fit three structural equation models: a cluster-robust model in long format, a two-level model that decomposes effects into within-dyad and between-dyad components, and a wide-format model with full equality constraints across dyad members. Then fit an unconstrained wide-format model and use a likelihood ratio test to assess whether the equality constraints are tenable. Repeat for all three outcomes.

Reflection prompt: For each outcome, do the likelihood ratio test results agree with the data-generating process (which has equal slopes across gender, with the single exception of `actor_sdt × gender_male` on engagement)? Which outcome's test result is the weakest signal in favour of indistinguishability, and why?

Section 4 — Distinguishable dyads: MLM moderator

What you are practising: adding `gender` as a moderator of actor and partner effects in a multilevel model.

What to produce: Load the long-format data. For each outcome, fit a baseline multilevel model (no gender interactions) and a full moderator model that interacts gender with every actor and partner predictor. Compare the two via a likelihood ratio test. For each outcome, use a simple-slopes tool to extract the actor effects (and partner effects, where relevant) separately for each gender, and report the profile-likelihood confidence intervals. Repeat for all three outcomes.

Reflection prompt: The data-generating process specifies that only `actor_sdt × gender_male` on engagement is non-zero. Do the simple-slope

plots and likelihood ratio tests pick this up cleanly, or do other interactions appear spuriously significant?

Section 5 — Distinguishable dyads: SEM (two-level with moderation)

What you are practising: mirroring the multilevel moderator approach in SEM with calculated simple slopes.

What to produce: Load the long-format data. Construct a numeric gender indicator and create manual interaction columns for every actor and partner predictor. For each outcome, specify a two-level SEM with the gender interactions at the within-dyad level and the three moderators at the between-dyad level. Add calculated simple-slope parameters (one per gender per pathway). Fit the model and compare the within-dyad SEM gender coefficients with the multilevel estimates from the previous section. Repeat for all three outcomes.

Reflection prompt: The within-dyad SEM gender coefficients should mirror the multilevel estimates from the previous section. Where do they differ, and why?

Section 6 — Distinguishable dyads: SEM wide format

What you are practising: the unconstrained / slopes-equal / fully-constrained model sequence and the three nested likelihood ratio tests.

What to produce: Load the wide-format data. For each outcome, fit three wide-format SEMs: one with all paths free to vary across dyad members, one that constrains slopes to be equal but leaves intercepts free, and one that constrains everything (slopes, intercepts, residual variances). Run the three nested likelihood ratio tests per outcome: slope equality, intercept equality, and overall indistinguishability. Build a fit-indices table that compares the three models for each outcome.

Reflection prompt: For which outcome is the slope-equality test most likely to be non-significant — and is that consistent with the single `actor_sdt × gender_male` interaction in the data-generating process?

Section 7 — Two-intercept models

What you are practising: the parameterisation trick that directly estimates separate intercepts per group, and its SEM analogue.

What to produce: Load the long-format data. For each outcome, fit three nested multilevel models: one with equal intercepts and equal slopes, one with separate intercepts for each gender but equal slopes, and one with separate intercepts and separate slopes. Use the parameterisation that suppresses the global intercept and replaces it with separate group intercepts. Compare adjacent models with likelihood ratio tests. Repeat for all three outcomes. Also fit the SEM analogue in wide format (two intercepts, equal slopes) and confirm the intercept estimates match the multilevel model.

Reflection prompt: The two-intercept approach is recommended when the researcher wants to report the absolute mean of each group (not the contrast with a reference). Which of the three outcomes has the most substantively different male vs. female mean, and is that consistent with what you saw in Section 1?

Section 8 — Moderated APIM (the centrepiece)

What you are practising: testing whether the partner crossover effect on each outcome is moderated by a dyad-level variable, using both multilevel modelling and SEM.

What to produce: Load both the long- and wide-format data. For each outcome, fit a baseline multilevel model and a moderated multilevel model that includes the focal partner-by-moderator interaction for that outcome. Use a simple-slopes tool to extract and visualise the partner predictor's effect at low and high levels of the focal moderator. Run the likelihood ratio test comparing base to moderated. Then for each outcome, build the corresponding wide-format SEM with a manually created product indicator and computed simple-slope parameters, and compare the simple-slope estimates with those from the multilevel model. Repeat for all three outcomes.

Reflection prompt: Compare the multilevel and SEM simple-slope estimates for each outcome. Where do they differ, and is the difference larger when the moderator is binary (`live_together`) or continuous (`years_together`, `time_spent_this_morning_together`)? Why might that be?

Section 9 — Integrative exercise

What you are practising: combining everything you have learned to answer a single, fully-specified research question.

What to produce: Pick one of the three outcomes. Build a kitchen-sink multilevel model that includes all three actor predictors, all three partner predictors, all three dyad-level moderators, the focal crossover-by-moderator interaction for

your chosen outcome, and a random intercept. Build the corresponding wide-format SEM with a manually created product indicator and computed simple-slope parameters at three levels of the moderator (low, mean, high). Report the actor and partner effect estimates with standard errors, the focal interaction coefficient from both approaches, the three simple slopes, and the SEM fit indices. Write a 4–6 sentence interpretation paragraph covering the substantive size and direction of the partner crossover effect, whether the moderator meaningfully changes that effect, a plain-language interpretation, and one limitation or follow-up question.

Reflection prompt: If you had to present this in a 5-minute conference talk, which figure and which one-sentence result would you lead with?